For a well-designed SAR ADC, quantization noise should typically dominate over other noise sources, of which thermal noise (kT/C) is the most significant. To maximize the performance of the ADC and preserve the Effective Number of Bits (ENOB), we must ensure that $kT/C \ll P_q$ (Quantization Noise Power).

Note: For these derivations and calculations, a single/differential-ended ADC models are used as required.

1. Thermal Noise Model

The noise power introduced across the DAC capacitor by the sampling resistor can be modeled as:

$$P_{tn} = 4kTR \cdot B$$

Where R is the total noise-generating resistance, and B is the noise equivalent bandwidth. Due to the RC low-pass filter nature of the sampling network, the noise equivalent bandwidth is:

$$B = \frac{1}{2\pi RC}$$

While a rough estimate using the bandwidth yields $\frac{2kT}{\pi C}$, integrating the noise power across the RC network over the entire frequency range yields the exact total thermal noise power:

$$P_{tn} = \frac{kT}{C}$$

If we set an upper limit to kT/C, then the total sampling thermal noise will be bounded. Hereafter, we will refer to this thermal noise power as $P_{kt/c} = kT/C$.

2. SNR and ENOB Degradation

For an ideal quantizer, the Signal-to-Noise Ratio (SNR) in dB is governed by:

$$\mathrm{SNR}_{ideal}(\mathrm{dB}) \approx 6.02 \cdot N + 1.76$$

(where N is the number of bits)

When other noise sources are considered, the overall SNR decreases, resulting in a lower ENOB:

$$\mathrm{SNR}(\mathrm{dB}) = 10 \log_{10}\left(\frac{P_{sig}}{P_{tot}}\right)$$

For an ideal quantizer, $P_{tot} = P_q$. Now, assume additional noise sources are introduced and can be represented as a scaled factor a of P_q:

$$P_{other} = P_{kt/c} + P_{comp} + P_{switch} = a \cdot P_q$$

$$P_{tot} = P_q + a \cdot P_q = (1 + a) \cdot P_q$$

The degraded SNR and ENOB reduction are then:

$$\mathrm{SNR}_{degraded}(\mathrm{dB}) = \mathrm{SNR}_{ideal}(\mathrm{dB}) - 10\log_{10}(1 + a)$$

$$\mathrm{ENOB}_{reduction} = \frac{10\log_{10}(1 + a)}{6.02} \approx 1.66\log_{10}(1 + a)$$

a (Noise Ratio)	ENOB Reduction (Bits)
0.01	0.007
0.10	0.070
0.20	0.130
0.40	0.240
0.50	0.290
1.00	0.500
2.00	0.790
3.00	1.000

3. Quantization Noise Power (P_q)(Differential Ended)

$$P_q = \frac{\mathrm{LSB}^2}{12} = \frac{\left(\frac{2 \cdot V_{ref}}{2^N}\right)^2}{12} = \frac{4 \cdot V_{ref}^2}{12 \cdot 2^{2N}} = \frac{V_{ref}^2}{3 \cdot 2^{2N}}$$

For our specific design ($V_{ref} = 1.0\ \mathrm{V}$, $N = 8$ bits):

$$P_q = \frac{1^2}{3 \cdot 2^{16}} = \frac{1}{196608} \approx 5.086 \times 10^{-6}\ \mathrm{V}^2$$

4. Capacitor Sizing for Thermal Noise

Assuming the dominant extra noise source is kT/C (so $a \approx A$), considering the maximum operating temperature of 125°C (398.15 K) ensures the inequality holds for all lower temperatures:

$$\frac{kT}{C} < a \cdot 5.086 \times 10^{-6}\ \mathrm{V}^2$$

$$\frac{5.494 \times 10^{-21}}{C} < a \cdot 5.086 \times 10^{-6}$$

$$C > \frac{1.080 \times 10^{-15}}{a}\ (\mathrm{F})$$

a (Noise Ratio)	Min Total C_DAC	Min Unit Cap (C_DAC / 2^8)
0.01	108.0 fF	0.422 fF
0.10	10.8 fF	0.042 fF
1.00	1.1 fF	0.004 fF

Conclusion: *Quantitative analysis shows that the minimum capacitance required to meet thermal noise constraints is significantly smaller than practical, manufacturable capacitor sizes for an 8-bit architecture. Therefore, capacitor sizing for this ADC is not limited by kT/C noise, but rather bounded by matching limitations, settling time, and parasitic considerations. We can also use a statistic-based approach for the noise model, but the deterministic derivations above are sufficient to loosen the noise-based size restrictions.*

5. Capacitor Array Mismatch Analysis (Pelgrom's Model)

To avoid linearity degradation (INL/DNL), matching between adjacent unit capacitors (C_u) must strictly adhere to Pelgrom's mismatch model:

$$\sigma\left(\frac{\Delta C}{C}\right) = \frac{A_c}{\sqrt{\mathrm{Area}}}$$

5.1 SPICE Model Extraction for Sky130 MiM Capacitors

We target the Metal-Insulator-Metal (MiM) capacitor devices available in the `sky130_fd_pr` library (`sky130_fd_pr__cap_mim_m3_1` / `_2`). The PDK SPICE model defines matching distributions as follows:

```
.param carea = 'camimc*(wc)*(lc)'
.param cperim = 'cpmimc*((wc)+(lc))*2'
.param czero = 'carea + cperim' dev/gauss='0.01*2.8*(carea + cperim)/sqrt(wc*lc*mf)'
```

Extracting the Gaussian deviation parameter yields the Pelgrom area coefficient: $A_c = 0.01 \cdot 2.8 = 0.028$ (or $2.8\% \cdot \mu\mathrm{m}$).

- **Area Capacitance (C_{area})**: $2.0\ \mathrm{fF}/\mu\mathrm{m}^2$
- **Perimeter Capacitance (C_{perim})**: $0.19\ \mathrm{fF}/\mu\mathrm{m}$

5.2 Mathematical Grounding: Stochastic Analysis of the DAC

The output of a capacitive DAC is defined by the ratio of the switched capacitors to the total capacitance. In a fully differential architecture, the differential output V_{diff} is:

$$V_{diff} = (V_{P+} - V_{P-}) - (V_{N+} - V_{N-})$$

The Unit Capacitor Model

Each unit capacitor C_i is modeled as an independent and identically distributed (i.i.d.) random variable: $C_i \sim \mathcal{N}(\mu_C, \sigma_C^2)$. The relative standard deviation (coefficient of variation) is defined by Pelgrom's Law:

$$\frac{\sigma_C}{\mu_C} = \frac{A_C}{\sqrt{W \cdot L}}$$

Variance of a Binary Weighted Bit

The k-th bit capacitor C_k is the sum of $n = 2^{k-1}$ unit capacitors. By the property of the sum of independent normal variables:

$$Var(C_k) = \sum_{i=1}^{2^{k-1}} Var(C_u) = 2^{k-1} \cdot \sigma_u^2$$

The Differential DNL at Mid-Scale

The Differential Non-Linearity (DNL) is worst at the MSB transition ($0\ldots \rightarrow 1\ldots$). The error E_{DNL} is the difference between the MSB capacitor and the sum of all lower-order capacitors (C_{rem}):

$$E_{DNL} = C_{MSB} - C_{rem}$$

For an 8-bit ADC, C_{MSB} has $2^7 = 128$ units, and C_{rem} also has 128 units. In a single-ended setup:

$$Var(E_{DNL,single}) = Var(C_{128}) + Var(C_{128}) = 256 \cdot \sigma_u^2$$

In a fully differential system, we have two such arrays (P and N) acting in opposition. The total differential error variance is:

$$Var(E_{DNL,diff}) = Var(E_{DNL,P}) + Var(E_{DNL,N}) = 256\sigma_u^2 + 256\sigma_u^2 = 512\sigma_u^2$$

$$\sigma_{DNL,diff} = \sqrt{512} \cdot \sigma_u = 2^{(N+1)/2} \cdot \sigma_u$$

5.3 Derivation of the Sizing Constraint

To ensure DNL < 0.5 LSB with a 3σ yield (99.7%), we set the boundary condition: $3 \cdot \sigma_{DNL,diff} < 0.5 \cdot \mu_C$. Substituting the differential variance:

$$3 \cdot \sqrt{512} \cdot \sigma_u < 0.5 \cdot \mu_C$$

$$67.88 \cdot \sigma_u < 0.5 \cdot \mu_C \implies \frac{\sigma_u}{\mu_C} < \frac{0.5}{67.88} \approx 0.007366$$

5.4 Physical Sizing for SKY130

Given the PDK parameters: $A_C = 0.028\ \mu\text{m}$, $C_{area} = 2\ \text{fF}/\mu\text{m}^2$, and $C_{perim} = 0.19\ \text{fF}/\mu\text{m}$.

Step 1: Solving for Minimum Area

Using Pelgrom's Equation:

$$\sqrt{\text{Area}} > \frac{A_C}{0.007366} = \frac{0.028}{0.007366} \approx 3.801\ \mu\text{m} \implies \text{Area}_{min} \approx 14.45\ \mu\text{m}^2$$

Step 2: Determining Dimensions (Square Geometry)

For a square capacitor to minimize fringe field asymmetry:

$$W = L = \sqrt{14.45} \approx \mathbf{3.8\ \mu m}$$

Step 3: Final Unit Capacitance (C_u) Calculation

$$C_{total} = (14.45 \cdot 2.0) + (4 \cdot 3.8 \cdot 0.19) = 28.9 + 2.888 = \mathbf{31.79\ fF}$$

5.5 Summary for Technical Report

Parameter	Symbol	Mathematical Derivation	Value
Max Allowable Sigma	σ_u/μ_C	$\frac{0.5}{3 \cdot \sqrt{2^{N+1}}}$	0.736%
Required Area	A	$(A_C/\sigma_{rel})^2$	14.45 µm²
Unit Dimensions	W, L	$\sqrt{A}$	3.8 µm × 3.8 µm
Unit Capacitance	C_u	$A \cdot C_{area} + P \cdot C_{perim}$	31.8 fF
Total Array Cap (per side)	C_{tot}	$2^8 \cdot C_u$	8.14 pF

5.6 Practical Unit Capacitor Selection

The calculations in the previous section establish the absolute lower limit for the unit capacitor ($W = L = 3.8\ \mu\text{m}$, $C_u \approx 31.8\ \text{fF}$) to satisfy stochastic matching constraints. For layout simplicity and to provide an additional margin for matching, we evaluate slightly larger, integer-dimensioned geometries:

- **Option A:** $4\ \mu\text{m} \times 4\ \mu\text{m}$
  - Area = $16\ \mu\text{m}^2$, Perimeter = $16\ \mu\text{m}$
  - $C_u = (16 \cdot 2.0) + (16 \cdot 0.19) = 32.0 + 3.04 = \mathbf{35.04\ fF}$
- **Option B:** $5\ \mu\text{m} \times 5\ \mu\text{m}$
  - Area = $25\ \mu\text{m}^2$, Perimeter = $20\ \mu\text{m}$
  - $C_u = (25 \cdot 2.0) + (20 \cdot 0.19) = 50.0 + 3.8 = \mathbf{53.8\ fF}$

We will choose one of these two practical unit sizes for the final layout implementation based on the final routing density and available area.

6. Architecture Implications and Area Feasibility

A standard binary-weighted Capacitor DAC for an 8-bit differential architecture requires 512 total unit capacitors (256 per side).

$$\text{Total Active Cap Area} = 512 \cdot 14.45\ \mu\text{m}^2 \approx 7,398.4\ \mu\text{m}^2$$

- Required Area (Array Only): **~7,400 µm²**
- Available Area (Total 1x2 Tiles): **36,000 µm² (160 x 225 µm)**

Unlike previous worst-case approximations, this stochastic analysis proves that implementing a standard full-binary CDAC is highly feasible within the allocated TinyTapeout area constraints without requiring complex area-reduction strategies.

7. References

- Hsu, C. P. (2013). "Analysis of capacitor mismatch in SAR ADC." *IEEE Trans. on Circuits and Systems*.
- Jan Henning Mueller, Sebastian Strache, Laurens Busch, Ralf Wunderlich, and Stefan Heinen. "The Impact of Noise and Mismatch on SAR ADCs and a Calibratable Capacitance Array Based Approach for High Resolutions." *INTL JOURNAL OF ELECTRONICS AND TELECOMMUNICATIONS*, 2013, VOL. 59, NO. 2, PP. 161–167. DOI: 10.2478/eletel-2013-0019